Corner-Adapted Quadrature for Rellich Normalization

Why the current rule fails at reentrant corners, and what replaces it

lappy development notes

Abstract

The Rellich identity computes $\|u\|_{L^2(\Omega)}$ from boundary Cauchy data alone. Its integrand carries a factor $r^{2\nu-2}$ at a corner of interior angle α, where $\nu = \pi/\alpha$; at a reentrant corner $\nu < 1$ and this is singular. We show (i) why the singularity can be removed *geometrically* by placing the Rellich reference point x_0 at the corner, (ii) why the current graded rule cannot integrate it when x_0 is placed elsewhere — the failure is truncation, not under-resolution — and (iii) that a corner-local Gauss–Jacobi rule in the variable $t = r^\sigma$ integrates it to machine precision with $O(10)$ nodes when σ rationalizes the corner's exponent family, and at a high-order algebraic rate otherwise — removing the constraint on x_0 in either case.

Read Section 7 before implementing. Sections 1–6 are the original analysis, and its central claim survived; three of its subsidiary conclusions did not. The substitution that rationalizes the exponent family is $t = r^{1/q}$ rather than $t = r^\nu$, and is unusable in that form for $q \geq 4$; the $2/\nu \in \mathbb{Z}$ condition is a limitation of $t = r^\nu$ specifically, not of the method; and the recipe of Section 6 omits a geometric cap on panel length whose absence costs twelve orders on some domains.

1 The identity and its corner integrand

For $-\Delta u = \lambda u$ in $\Omega \subset \mathbb{R}^2$ with $u = 0$ on $\partial\Omega$, the Rellich–Pohozaev identity with reference point $x_0 \in \mathbb{C}$ gives the L^2 norm as a pure boundary integral,

$$\|u\|^2_{L^2(\Omega)} \;=\; \frac{1}{2\lambda}\int_{\partial\Omega} \big((x - x_0)\cdot n\big)\Big(\frac{\partial u}{\partial n}\Big)^2 \mathrm{d}s \;=\; \frac{1}{2\lambda}\int_{\partial\Omega} r_N \Big(\frac{\partial u}{\partial n}\Big)^2 \mathrm{d}s, \tag{1}$$

with $r_N := (x - x_0)\cdot n$. The identity is exact for *every* x_0; the choice of x_0 affects only how hard the integral is to evaluate.

1.1 Local structure at a corner

Let the corner sit at the origin with interior angle α and edges along $\theta = 0$ and $\theta = \alpha$, and set

$$\nu \;:=\; \frac{\pi}{\alpha}, \qquad \nu_k := k\nu = \frac{k\pi}{\alpha}, \quad k = 1, 2, \ldots \tag{2}$$

The corner is *reentrant* when $\alpha > \pi$, i.e. $\nu < 1$; the 270° corner has $\nu = 2/3$ and the slit limit $\alpha \to 2\pi$ gives $\nu \to 1/2$. Any Dirichlet solution admits the local expansion

$$u(r,\theta) \;=\; \sum_{k\geq 1} c_k\, J_{\nu_k}\big(\sqrt{\lambda}\, r\big)\, \sin(\nu_k\theta). \tag{3}$$

On the edge $\theta = 0$ the outward normal is $n = -e_\theta$, so $u \equiv 0$ there forces the normal derivative to come entirely from the angular derivative:

$$\frac{\partial u}{\partial n}\bigg|_{\theta=0} = -\frac{1}{r}\frac{\partial u}{\partial \theta}\bigg|_{\theta=0} = -\sum_{k\geq 1} c_k\, \nu_k\, \frac{J_{\nu_k}(\sqrt{\lambda} r)}{r}. \tag{4}$$

Using $J_\mu(z) = (z/2)^\mu A_\mu(z^2)$ with A_μ entire and $A_\mu(0) \neq 0$,

$$\frac{\partial u}{\partial n}\bigg|_{\theta=0} = r^{\nu-1} F(r), \qquad F(r) = \sum_{k\geq 1} b_k\, r^{(k-1)\nu} A_{\nu_k}(\lambda r^2), \tag{5}$$

so that

$$\left(\frac{\partial u}{\partial n}\right)^2\bigg|_{\theta=0} = r^{2\nu-2}\, G(r), \qquad G(r) \;=\; F(r)^2 \;=\; \sum_{p\geq 0}\sum_{q\geq 0} g_{pq}\, r^{\,p\nu+2q}. \tag{6}$$

Two families of exponents appear in G: the *corner* exponents $r^{p\nu}$ generated by the expansion (3), and the *Bessel* exponents r^{2q} from the entire factors. This double structure is the crux of Section 3.1.

1.2 The geometric escape: $r_N \equiv 0$

Proposition 1. *If x_0 is placed at the corner, then $r_N \equiv 0$ on both straight edges emanating from it.*

Proof. On the edge $\theta = 0$, $x - x_0 = re^{i\cdot 0}$ is parallel to the edge, while $n = -e_\theta$ is perpendicular to it; hence $(x - x_0)\cdot n = 0$ pointwise, and likewise on $\theta = \alpha$. ☐

The singular factor $r^{2\nu-2}$ is therefore multiplied by an *exact* zero, and the corner contributes nothing: the boundary integral collapses onto the parts of $\partial\Omega$ where u is smooth. This is why placing x_0 at a reentrant corner yields machine precision with any reasonable quadrature. It is also why the escape does not generalize: x_0 is a single point, and a domain with several reentrant corners (H-shape, GWW) cannot zero r_N at all of them.

2 Why the graded rule fails for x_0 off the corner

For general x_0, r_N is a nonzero constant along each straight edge (it is the signed distance from x_0 to the edge's line), and the edge contributes

$$I_{\text{edge}} \;=\; \rho \int_0^R r^{2\nu-2} G(r)\,\mathrm{d}r, \qquad \rho := r_N = \text{const.} \tag{7}$$

This converges for $\nu > 1/2$, but the convergence is deceptive.

Lemma 1 (Mass concentration)**.** *The fraction of $\int_0^R r^{2\nu-2}\,\mathrm{d}r$ contributed by $r < \varepsilon$ is $(\varepsilon/R)^{2\nu-1}$.*

The exponent $2\nu - 1$ vanishes as $\nu \to 1/2$. Numerically, with $R = 1$ and $\varepsilon = 10^{-9}$:

α	ν	$2\nu-1$	mass below 10^{-9}
1.25π	0.800	0.600	4.0×10^{-6}
1.5π	0.667	0.333	1.0×10^{-3}
1.75π	0.571	0.143	5.2×10^{-2}
1.9π	0.526	0.053	0.336
1.99π	0.503	0.005	0.901

For a near-slit corner, *ninety percent of the integral lives within* 10^{-9} *of the corner.* lappy's Kress-graded rule clamps its innermost node at `quad._KRESS_TAU_FLOOR` $= 10^{-9}$ (a guard against a genuine defect: nodes built as $p_0 + \tau L\,\hat{d}$ in absolute coordinates round to exactly p_0 once τL falls below machine epsilon relative to $|p_0|$). The rule is thus *truncated*, not under-resolved: refining from 565 to 6790 nodes leaves the innermost node at 10^{-9} and the error unmoved. Observed errors track Lemma 1 over eight orders of magnitude, scaling linearly in $|x_0 - \text{apex}|$ with constant $\approx (10^{-9})^{2\nu-1}$.

Remark 1. The floor is a symptom of *coordinate representation*, not of floating-point resolution. The value $r = 10^{-30}$ is perfectly representable; it is $p_0 + \tau L\hat{d}$ that is not. A rule evaluated in corner-local (r,θ) never forms that difference. This distinction is what makes Section 3 possible.

3 A corner-local Gauss–Jacobi rule

Gauss–Jacobi quadrature with weight $(1-x)^a(1+x)^b$ integrates $\int_{-1}^{1}(1-x)^a(1+x)^b f(x)\,\mathrm{d}x$ exactly for f polynomial of degree $\le 2n-1$, and spectrally for f analytic. Mapping $r = R(1+x)/2$ and matching $b = \beta := 2\nu - 2$ to (7),

$$\int_0^R r^{2\nu-2} G(r)\,\mathrm{d}r \;=\; \left(\frac{R}{2}\right)^{2\nu-1} \sum_{i=1}^{n} w_i\, G(r_i), \qquad r_i = \tfrac{R}{2}(1+x_i). \tag{8}$$

Admissibility requires $\beta > -1$, i.e. $\nu > 1/2$: precisely the range of integrability, degenerating exactly at the slit.

Two properties matter as much as the accuracy:

1. **The nodes stay away from the corner.** The smallest Jacobi node sits at distance $O(n^{-2})$, not 10^{-9}: for $n = 16$ the innermost node is near 10^{-3}. The singularity is absorbed *analytically* into the weight rather than resolved by clustering, so neither the truncation floor nor the coordinate cancellation of Remark 1 ever arises.

2. **Only ν is needed.** $G(r) = (\partial_n u)^2 r^{2-2\nu}$ is formed numerically from black-box Cauchy data; no analytic knowledge of u is used. Since $\nu = \pi/\alpha$ comes exactly from the geometry, this is free — but it must be exact, see Section 4.

3.1 The multi-exponent obstruction and the substitution

Rule (8) is spectral only if G is analytic. By (6) it is not: G carries fractional powers $r^{p\nu}$. For a *single* mode ($c_k = 0$, $k \ge 2$) we have $G(r) = b_1^2 A_{\nu_1}(\lambda r^2)^2$, analytic in r^2, and (8) is spectral — but a single mode is not the situation of interest. For a general solution, convergence degrades to algebraic.

Proposition 2 (Rationalizing substitution)**.** *Under $t = r^\nu$, equivalently $r = t^{1/\nu}$, the edge integral becomes*

$$\int_0^R r^{2\nu-2} G(r)\,\mathrm{d}r \;=\; \frac{1}{\nu}\int_0^{R^\nu} t^{\,1-1/\nu}\,\widetilde{G}(t)\,\mathrm{d}t, \qquad \widetilde{G}(t) = \sum_{p,q} g_{pq}\,t^{\,p+2q/\nu}, \tag{9}$$

a Jacobi-weight integral with exponent $\beta' = 1 - 1/\nu$. The corner family $r^{p\nu}$ maps to the integer powers t^p.

Proof. $\mathrm{d}r = \frac{1}{\nu} t^{1/\nu-1}\,\mathrm{d}t$ and $r^{2\nu-2} = t^{2-2/\nu}$; the exponents add to $(2-2/\nu)+(1/\nu-1) = 1-1/\nu$. Substituting $r^{p\nu+2q} = t^{p+2q/\nu}$ into (6) gives $\widetilde{G}$. □

Admissibility $\beta' > -1$ again requires exactly $\nu > 1/2$.

The caveat. The substitution rationalizes the corner family completely, but maps the Bessel family $r^{2q} \mapsto t^{2q/\nu}$, which is integral only when $2/\nu \in \mathbb{Z}$ — equivalently $\alpha = m\pi/2$ for an integer m. The 270° corner is such a case:

$$\alpha = \tfrac{3}{2}\pi \implies \nu = \tfrac{2}{3} \implies 2/\nu = 3 \in \mathbb{Z}, \qquad \widetilde{G}(t) = \sum g_{pq} t^{\,p+3q} \text{ (a power series in } t). \tag{10}$$

For generic α the residual exponents $2q/\nu$ are fractional, but they are *high order*: the leading one is $2/\nu > 2$, so $\widetilde{G} \in C^2$ at worst and Gauss convergence is algebraic of high order rather than spectral. This is confirmed in Section 5(c): the machine precision at $n = 8$ is specific to $\alpha = \frac{3}{2}\pi$, while generic α converges at roughly two digits per node doubling — still reaching 10^{-12} by $n = 32$, which is cheap enough that the distinction is of no practical consequence.

4 Sensitivity to ν

The weight exponent must use ν at full precision. Perturbing $\nu \to \nu + \delta$ leaves a residual factor $r^{-2\delta}$ in the integrand, which is unbounded as $r \to 0$ for $\delta > 0$ and destroys the analyticity the rule depends on. Measured at $\alpha = 1.5\pi$ (true $\nu = 2/3$), mode $(1, 1)$:

ν used in β	28 nodes	192 nodes
0.6667 (exact)	-7.5×10^{-15}	-3.1×10^{-15}
0.667	6.6×10^{-5}	1.7×10^{-5}
0.600	-1.8×10^{-2}	-4.7×10^{-3}

A relative error of 3×10^{-4} in ν costs four digits and reduces convergence to algebraic. Since $\nu = \pi/\alpha$ is exact from the geometry this costs nothing — but it rules out tabulated, rounded, or margin-padded exponents of the kind used for grading orders.

5 Numerical verification

All errors are relative, against closed-form sector eigenfunctions of unit norm (exact L^2 norms, so the reference is analytic rather than quadrature-based).

(a) Single mode, 270° corner, x_0 at the failing default. Cauchy data treated as a black box; $2n_{\mathrm{edge}} + n_{\mathrm{arc}}$ nodes total.

nodes	Gauss–Jacobi	existing Kress rule
16	-7.1×10^{-9}	—
28	-7.5×10^{-15}	—
2264	—	2.2×10^{-7} (plateau)

(b) Multi-exponent corner data. A five-term expansion (3) times an analytic tail, integrated against an mpmath reference at 40 digits, at $\nu = 2/3$:

n	Jacobi in r	Jacobi in $t = r^{\nu}$
8	-4.6×10^{-3}	4.0×10^{-15}
16	-1.2×10^{-3}	7.3×10^{-15}
64	-7.2×10^{-5}	1.6×10^{-13}

Plain Jacobi in r is algebraic, as Section 3.1 predicts; the substituted rule is spectral and *degrades* slowly past $n \approx 32$ from roundoff amplification, so small n is both cheaper and more accurate.

(c) Generic α: is $\nu = 2/3$ special? The same five-term data, substituted rule, against an *exact* reference (see the warning below).

α	$2/\nu$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
1.5π	3.000 (int.)	2.0×10^{-15}	2.0×10^{-15}	-2.6×10^{-14}	1.7×10^{-13}
1.33π	2.667	2.8×10^{-8}	2.7×10^{-10}	2.7×10^{-12}	5.1×10^{-14}
1.25π	2.500	4.7×10^{-8}	5.1×10^{-10}	5.9×10^{-12}	5.0×10^{-14}
1.6π	3.200	-5.6×10^{-9}	-3.4×10^{-11}	-1.6×10^{-13}	-7.7×10^{-13}
1.7π	3.400	-5.2×10^{-9}	-2.7×10^{-11}	-3.4×10^{-13}	-9.8×10^{-13}
1.85π	3.700	-1.5×10^{-9}	-6.0×10^{-12}	1.9×10^{-14}	-2.5×10^{-12}
1.95π	3.900	-1.4×10^{-10}	-2.0×10^{-13}	-4.0×10^{-13}	-1.2×10^{-11}

Only $\alpha = \frac{3}{2}\pi$ is machine-exact at $n = 8$, confirming Section 3.1: the integer $2/\nu$ is genuinely special. Every other angle converges at $\approx$ two digits per doubling — high-order algebraic, exactly as predicted — and all reach 10^{-12} or better by $n = 32$. The mild degradation past $n \approx 32$ is roundoff, not truncation.

Warning: mpmath's `quad` is not a safe reference here. The endpoint behaviour $r^{2\nu-2} \to r^{-1}$ as $\nu \to \frac{1}{2}$ defeats `mp.quad` even at 40 digits. Measured against the exact termwise value $\int_0^1 r^a \,\mathrm{d}r = (a+1)^{-1}$, its error reaches -4.7×10^{-5} at $\alpha = 1.85\pi$ and -4.3×10^{-2} at $\alpha = 1.95\pi$. Using it as ground truth produces convincing but entirely spurious "plateaus" in the rule under test. Any reference for this integral should be closed-form.

A trap worth recording. At $\alpha = 3\pi/2$ the two edges carry $r_N = \operatorname{Im} x_0$ and $-\operatorname{Re} x_0$ against identical $(\partial_n u)^2$, so any x_0 on the diagonal $\operatorname{Re} x_0 = \operatorname{Im} x_0$ cancels the singular contribution *identically.* A test using such an x_0 silently measures the smooth part of the boundary only. Corner-quadrature experiments should always report the edge/arc split.

6 Recommended construction

For each corner needing the rule (Section 7.6 — *not* simply $\nu < 1$, which is what this section originally said), and each straight edge meeting it:

1. Work in corner-local coordinates (r, θ); never form $p_0 + \tau L\hat{d}$.
2. Panel the edge, substitute $t = r^\nu$ with $\nu = \pi/\alpha$ exact.
3. Apply Gauss–Jacobi with $\beta' = 1 - 1/\nu$ and $n = 8$–16 nodes.

Smooth portions of $\partial\Omega$ retain plain Gauss–Legendre. Edges meeting two singular corners are split, one panel per endpoint.

This removes the requirement that x_0 sit at a corner, and with it the obstruction to multi-corner domains: no partition of unity over corners is needed. The requirement $\nu > 1/2$ is not restrictive — it is exactly the condition for (7) to converge at all — but the slit limit $\nu \to 1/2$ remains delicate, both weights degenerating as $\beta, \beta' \to -1$.

Not yet verified. A real `FourierBesselBasis` through this rule (as opposed to closed-form or synthetic data), and a genuinely multi-corner domain.

7 Addendum: what implementation changed

The analysis above was written before the rule was built. Its core result held: a corner-adapted rule reaches 10^{-14} where the graded rule plateaus at 10^{-7}, with fewer nodes. What follows corrects the parts that did not survive contact with a real domain, and records what could only be learned by measuring. Everything here is implemented in `lappy.quad` and `lappy.eigfun_integrals`, and tested in `tests/test_eigfun_integrals.py`.

7.1 The substitution is $t = r^{1/q}$, and $t = r^\nu$ is a trap for curved edges

Section 3.1 identified the exponent family as $\{k\nu + 2q\}$ and concluded that $t = r^\nu$ rationalizes it when $2/\nu \in \mathbb{Z}$ — for reentrant α, only $\alpha = 3\pi/2$. That is correct *for a straight edge*, and it is the wrong family for a curved one.

On a curved edge two things change, both adding *integer* powers of arclength s. First, r_N is no longer constant: with x_0 at the corner it is $(\kappa_0/2)s^2 + O(s^3)$ — vanishing to second order rather than identically, so Proposition 1's escape degrades gracefully rather than failing. Second, $r = |x(s) - \text{corner}| = s - (\kappa_0^2/24)s^3 + \cdots$, so r is not arclength. The family becomes $\{k\nu + m\}$, $m \in \mathbb{Z}_{\geq 0}$, including *odd* powers. Under $t = r^\nu$ that maps to $\{k + m/\nu\}$ whose leading residual $t^{1/\nu}$ has $1/\nu \in (1, 2)$: only C^1. Measured, it never reaches 10^{-13} at any order up to 64. Under $t = r^{1/q}$ with $\nu = p/q$ in lowest terms it maps to $\{kp + mq\}$ — all integers — and is exact by order 6–14.

But $t = r^{1/q}$ is unusable as written. Since $\tau = t^q$, the innermost node behaves as $\tau_{\min} \sim t^q_{\min}$:

α	q	$\tau_{\min}$ at order 24	
$3\pi/2$	3	1.4×10^{-8}	
$5\pi/4$	5	1.1×10^{-10}	(below the floor)
$7\pi/4$	7	4.7×10^{-19}	
$11\pi/6$	11	1.6×10^{-29}	

Those nodes round onto the corner once mapped through a segment's parametrization — the very defect Remark 1 identifies. The rule is exact as an abstract statement about $[0, 1]$ and worthless on a boundary.

7.2 What works: exactness from the weights, node placement from ν

The resolution separates the two roles the substitution was doing. Nodes are placed by $t = r^{\nu}$, whose crowding is mild ($\tau_{\min}$ stays at 10^{-5}–10^{-7}). Exactness comes from the *weights*: the exponent set $\{\gamma + j\nu + m\}$ is known, and every moment is closed-form, $\int_0^1 t^e \, \mathrm{d}t = (e + 1)^{-1}$, so the weights are obtained by solving a linear system. This also covers irrational ν, which the substitution cannot touch at all — and irrational ν is the *generic* case for a curved corner, whose angle is fixed by the geometry rather than chosen (two overlapping circles give $\nu = 0.657360$).

One trap: solving the square system (as many exponents as nodes) is exact on its span but has cond $\sim 10^{13}$ at order 12 and 10^{19} at order 16, with weights reaching -10^4. Since $\sum |w|$ multiplies every roundoff error in the integrand, that imports a 10^{-12} floor. Using *fewer exponents than nodes* — a minimum-norm least-squares solve — keeps $\sum |w| = 1.0$ with exactness intact.

7.3 The edge type matters, and conflating them costs seven orders

A straight edge admits only *even* integer powers (r_N exactly constant, r exactly arclength), so $\{\gamma + j\nu + 2q\}$; odd powers are curved-only. Scored against the curved set, the substitution rule at $\alpha = 3\pi/2$ reads 10^{-8} — against a measured 2.4×10^{-15} on a real sector. Any error indicator must use the family belonging to the edge it is scoring.

7.4 Panel length must be capped by clearance, not by a fraction

Section 6's recipe panels the edge but says nothing about how long a panel may be. The corner expansion is valid only within the largest disk about the corner inside Ω; a panel reaching past it is under-resolved, because the rule clusters its nodes *at* the corner and cannot resolve the $\sqrt{\lambda}$ oscillation over the remaining arclength. Measured on a $1 \times N$ polyomino strip with one cell below an end (so the corner's edge has length $N - 1$ against a clearance of 1), against its exact eigenfunction:

N	no cap	$0.9\times$ clearance
2	6.5×10^{-14}	5.3×10^{-15}
8	2.3×10^{-2}	1.6×10^{-15}
16	7.7×10^{-2}	1.3×10^{-15}
24	7.4×10^{-3}	4.4×10^{-16}

Twelve orders. A fixed *fraction* of the edge cannot substitute — being a fraction of the edge, it grows with the edge and stays too long (0.25 of the edge still gives 2.6×10^{-4} at $N = 16$). The cap must be geometric.

7.5 Two properties that look like bugs and are not

$\sum_i w_i = 1$ holds only when the constant function lies in the rule's exact class. Renormalizing would destroy exactness on the singular class the rule exists for; $\sum_i w_i - 1$ is instead its cheapest self-diagnostic. And accuracy is *not monotone* in order: past a ν-dependent threshold the integrand's dynamic range ($\tau^{2\nu-2} \sim 6 \times 10^8$ at $\tau \sim 10^{-9}$ against a correspondingly tiny weight) makes roundoff dominate. At $\nu = 0.526$ the best order is 16 while the floor-based cap is 75, and using the cap gives 4.2×10^{-9}. Order selection must therefore scan a calibrated curve, not increase the order until a target is met.

7.6 The criterion is not $\nu < 1$

Every section above frames the problem around *reentrant* corners, and the implementation followed: a corner got the adapted rule when $\nu < 1$, on the reasoning that a convex corner's eigenfunction is bounded and so a smooth rule is already exact. That reasoning is wrong, and it cost the suite several orders on domains with no reentrant corner at all.

What (6) actually says is that the integrand carries $r^{2\nu-2}$, and nothing in its derivation assumed $\nu < 1$. The relevant question is therefore not whether $2\nu - 2$ is negative but whether it is an *integer*. At $\alpha = \frac{3}{4}\pi$ we have $\nu = 4/3$: the eigenfunction is bounded, its normal derivative $r^{1/3}$ vanishes at the corner, and the integrand is $r^{2/3}$ — continuous, with an infinite derivative, and Gauss–Legendre converges on it algebraically. The analytic model integrand used to *size* a smooth panel cannot see this at all, so the rule reports success while missing by orders.

The test that works is direct: ask what an n-point Gauss–Legendre rule does to $\int_0^1 \tau^\gamma \, d\tau$ with $\gamma = 2\nu - 2$ (`quad.smooth_power_error`).

$\gamma = 2\nu - 2$	ν	$n = 16$	$n = 64$
0.333	1.167	8.0×10^{-5}	2.1×10^{-6}
0.667	1.333	1.2×10^{-5}	1.2×10^{-7}
1.000	1.500	0	0
2.500	2.250	1.7×10^{-9}	1.2×10^{-13}
11.550	6.775	8.7×10^{-16}	3.5×10^{-16}

Integer γ is a polynomial and integrates exactly, so $\nu = 3/2$ or 2 needs nothing. Large fractional γ has so many continuous derivatives that the algebraic rate is indistinguishable from spectral at any usable order. Only small fractional powers — corners near 135° — genuinely defeat a smooth rule.

Both directions matter. A naive "ν is not an integer" criterion is also wrong: it hands the adapted rule to $\nu = 6.78$ corners that do not need it and makes them *worse* ($1.2 \times 10^{-11} \to 8.9 \times 10^{-10}$, measured on `iso_tri_h05`), because the rule is not exact on constants and the integrand there has no singular structure to exploit. Asking the smooth rule what it can deliver settles both cases. Measured end to end, by refining the node set and differencing the integral:

domain	nodes	before	after
`right_trapezoid`	$84 \to 212$	1.8×10^{-6}	2.9×10^{-16}
`GWW1`	$204 \to 440$	3.2×10^{-5}	5.0×10^{-13}
`reg_ngon_7`	$98 \to 868$	2.6×10^{-5}	1.6×10^{-13}
`iso_tri_h05`	$96 \to 224$	1.2×10^{-11}	4.8×10^{-16}

7.7 On a curved boundary the parametrization can cost more than the eigenfunction

The analysis above sizes panels for the eigenfunction: its corner exponents and its $\sqrt{\lambda}$ oscillation. On a curved segment that is not the whole cost. Panels live in normalized arclength, and for a curve of varying speed that map is analytic but with a strip of analyticity that narrows as the curve becomes eccentric — taking Gauss–Legendre's rate with it.

This is measurable with no eigenfunction at all. By the divergence theorem $\int_{\partial\Omega} r_N \, ds = 2|\Omega|$ for any x_0, a closed form containing no λ, no basis and no solution:

order	16	32	64	128	192	256
disk	3e−16	3e−16	3e−16	3e−16	3e−16	3e−16
ellipse $a = 2$	5e−03	2e−04	4e−07	3e−12	5e−15	2e−15
ellipse $a = 4$	4e−02	1e−02	2e−03	8e−05	3e−06	1e−07
ellipse $a = 6$	6e−02	3e−02	1e−02	2e−03	5e−04	1e−04

A circle is exact at any order because its arclength map is linear; so is a straight edge. Sizing each segment on this identity as well as on the oscillation (`eigfun_integrals.geometry_order_for_precision`) costs an $a = 2$ ellipse $46 \to 168$ nodes and takes its eigenfunction integrals from 1.0×10^{-6} to 3.9×10^{-16}, while every exactly-parametrized boundary pays nothing.

This is *not* the arclength table's interpolation tolerance, which was the natural suspect: the error is identical to three digits at tol $= 10^{-4}$, 10^{-6} and 10^{-8}, while the table's build cost goes 0.0s, 0.9s, 65s. Refining it is pure expense.

Past about $a = 4$ no usable order suffices. That is reported rather than chased: `BoundaryQuad.precision` stops claiming 10^{-13}, `shortfalls` names the segment, and the warning says the parametrization is the limit so the reader does not go looking for a defect in the eigenfunction. The same applies to a corner too close to the slit limit to be admissible, where `precision` is ∞.

A third class remains open and is labelled rather than fixed: a `stadium` has exact parametrization and integer ν, yet its integrals sit at 1.5×10^{-4} because its boundary data carries far more harmonics than $2\sqrt{\lambda}\,L$ predicts. No geometry-only rule can know that — it is a property of the solution. Measured against the Cauchy data's own spectrum the model errs in both directions: a circle needs 4 harmonics where the model assumes 64, an ellipse needs 80 where it assumes 8.

7.8 Scope, sharpened

The rule is matched to *eigenfunction*-level Cauchy data, whose local structure is purely the corner family. It is *not* appropriate for a basis-level $N \times N$ Gram matrix: columns centred at other corners are plain analytic at this one, with $O(1)$ amplitude at every corner simultaneously. Measured, the corner rule beats the graded rule by 3–8 orders on corner-family blocks and loses by 2–4 on mixed ones. That is why `lappy` retired the basis-level path rather than adapting it.

Finally, a warning about instruments. `mpmath.quad` is not a valid reference for these integrands: as $\nu \to 1/2$ the endpoint behaviour approaches r^{-1} and it errs by 4×10^{-2} even at 40 digits, manufacturing convincing but entirely spurious convergence plateaus. Every reference used in the tests is closed-form.

Nor is x_0-invariance an error bar, though it is a fine detector. The identity holds for every x_0, so disagreement across x_0 is real — it is what exposed the shortfall of Section 7.6 in the first place — but it does not measure the quadrature alone. It also carries the *eigenvalue's* error, since λ enters through $1/(2\lambda)$ and through the Cauchy data, so an inexact λ makes u not quite an eigenfunction and the x_0-dependence reappears. On `L_shape` the spread reads 3.3×10^{-12} where the quadrature is 1.4×10^{-13}; on `H_shape` it reads 4.1×10^{-11} where the quadrature is 1.0×10^{-8}, understating it by three orders. Refining the node set and differencing the integral (`eigfun_integrals.verify_gram`) measures the quadrature and nothing else, and is the instrument to reach for.

It is also worth recording what the spread is *not*: it is not dominated by the eigenfunction's boundary residual. Holding the node set fixed and sweeping the basis size $60 \to 480$ moves $\sup |u|$ on the boundary by six orders ($9.3 \times 10^{-10} \to 1.0 \times 10^{-15}$) and leaves the spread at 1.68×10^{-6}, identical to three significant figures.